

There would appear to be two reasons why not.

antecedently extremely implausible. So even if this piece of evidence favors it, it is

First: the hypothesis that Satan controls the weather in South Bend is

very implausible.

just making a wildly implausible hypothesis just a wildly implausible — but still

$$P(a|b) = \frac{P(a \& b)}{P(b)}$$

To go back to the main lecture, click [here](#).

Derivation of Bayes' theorem

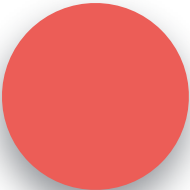
1. $P(a b) = \frac{P(a \& b)}{P(b)}$	def. of conditional probability
2. $P(b a) = \frac{P(a \& b)}{P(a)}$	def. of conditional probability
3. $P(a b) * P(b) = P(a \& b)$	(1), multiplication by '='s
4. $P(a \& b) = P(b a) * P(a)$	(2), multiplication by '='s
5. $P(a b) * P(b) = P(b a) * P(a)$	(3),(4)
C. $P(a b) = \frac{P(b a) * P(a)}{P(b)}$	(5), division by '='s

Remember our definition of conditional probability:

Using this definition, we can prove Bayes' theorem as follows:

Appendix: Proof of Bayes' theorem









follows.

contemporary physics which will be important to the argument which

Enough about probability (for now). Let's turn now to some results from

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensional constants' will be our focus.

energy density of empty space.

One such constant is the cosmological constant, which measures the

The empirical constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how

likely it is, given the laws of the nature, that the fundamental constants

(like the cosmological constant) would fall in a certain range.

life to have evolved. For example, if there were nothing but hydrogen, it is

hard to see how life could have evolved. If there were no planets, it is hard

We can make certain plausible assumptions about what it would take for

to see how life could have evolved.

constant has a value which would permit the evolution of life.

Given assumptions such as these, we can look at what the standard model

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.



cosmological constant falling in a life-supporting range, giving the laws of

The claim is that, according to current physics, the probability of the

nature, is

1



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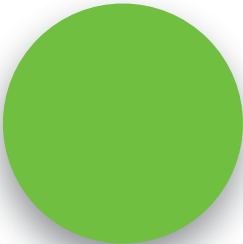
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fine-
tuning of the
universe

A solid red circle serves as the background for the text.

Bayes'
theorem

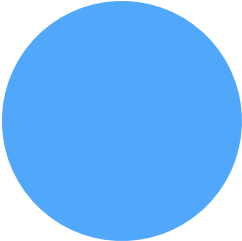








the
argument





The claim is that, according to current physics, the probability of the

completing a constant falling in a life-supporting range, given the laws of

nature, is

1

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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and



of the universe.

Much as we considered two different hypotheses about the number of balls

in the turn, so we can consider two different hypotheses about the origins

created by an intelligent designer. Let the non-design hypothesis be the

hypothesis that it was not.

Let the design hypothesis be the hypothesis that the universe

value in the life-permitting range.

Let e be the fact that the cosmological constant has a

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e|\text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the non-

numbers.

design hypothesis given evidence is that just plugging in the



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$



consider the odds of winning Powerball one trillion times in a row. Call that a “super

PowerBar!

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

It is difficult to think about numbers as large as the denominator of this fraction. But

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

Now consider the odds of winning a super Powerball one trillion times in a row.

odd of the universe being life-permitting by chance.

This means that if you simply take the physics at face value, and begin by assigning a

superduperpowerball a trillion times in a row.

probability of 0.5 to the non-design hypothesis, you should think that the chances of

the non-design hypothesis being true are vastly lower than the chances of winning a

minus the very small number we're just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.







2



P

$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$



6. *Life exists.*

1, 2, 3, 4

$$\text{C.P.}(\text{God exists}) \approx 1. (7,8)$$

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

8. If the universe was created by an

intelligent designer, then God exists.

5



P

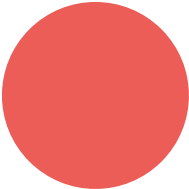
4. Bayes', theorem.

$$(life/death) \equiv 0.5$$

$$(1\text{ife} \mid \text{non-design}) = \frac{1}{10^{120}}$$

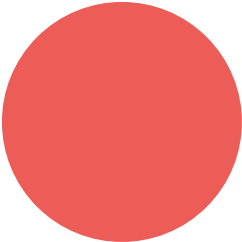
$$(\text{design} / \text{life}) \approx 1$$







fine-
tuning of the
universe



it. You know that the jar either has 10 ping pong balls in it or 1000 ping

Suppose that have an opaque jar with some number of ping pong balls in

think that the two possibilities are equally likely.

pong balls in it. In either case the balls are numbered sequentially. You

Suppose that we pick a ball at random. It is #3. Does this favor one

hypothesis over the other?

hypothesis makes it more likely that #3 will be picked (rather than

some number higher than 10).

It seems to favor the 10 ping pong ball hypothesis, since that

Suppose someone offered you 40:1 odds that it is the 10-ball jar — so

you win \$40 if it is the 1000 ball jar, and lose \$1 if it is the 10 ball jar.

Should you take the bet?

But how much? How likely should I think that the jar has 10

ping pong balls in it, vs. 10000?



will tell us how to calculate probabilities in this kind of case.

hypotheses: the 10-ball theory and the 1000-ball theory.

What information do we have? We are comparing two different

probability of being correct. This is called the theory's prior

Before we drew the ball, we thought that each theory had a 0.5

probability.

We also know something else. We know likely each theory

predicts it to be that we draw the #3 ball.

says that there is 0.001 probability that this ball will be drawn.

We know that the 10-ball theory says that there is a 0.1

probability that the #3 ball will be drawn. The 1000-ball theory

is the probability that something will happen if something else is

This is a claim about what is called conditional probability. This

evidence (e) conditional on the hypothesis (h) . We write that like

true. What we see in the probability of the

this:

Pr(e|h)

probability of the evidence given the hypothesis. What we want to

known is this:

So we know the prior probability of the hypothesis, and the

Pr(h|e)

us how likely we should take the hypothesis to be, given the

evidence never observed.

This is the probability of the hypothesis given the evidence. It tells

How can we do this?

One new way to answer these questions employs a

mini nister.

century English mathematician and Presbyterian

widely accepted rule of reasoning called "Bayes'

theorem," named after Thomas Bayes, an 18th

$$P(a|b) = \frac{P(a \& b)}{P(b)}$$